

Project Blueprint & Work Division Plan

Flood Evacuation Time Modeling via Partial Differentiation

PESCE Coding Club — Social Impact Track

September 16, 2026

1 Project Overview

This project models and analyzes flood evacuation times (U) using partial derivatives in two variables:

- x : Population Density
- y : Number of Open Evacuation Routes

The goal is to analyze real-world/historical flood data via MATLAB to evaluate sensitivities ($\frac{\partial U}{\partial x}$ and $\frac{\partial U}{\partial y}$) for disaster-response planning.

2 Team Work Division

Member	Primary Domain	Key Responsibilities
Member 1	MATLAB & Math	• Formulate $U(x, y)$ mathematical function. • Write MATLAB code to import spreadsheet data. • Calculate $\frac{\partial U}{\partial x}$ and $\frac{\partial U}{\partial y}$ symbolically & numerically. • Export derivative calculations to output spreadsheet.
Member 2	Data & Report	• Research 2–3 years of historical flood data. • Build and structure <code>flood_data.xlsx</code>. • Draft final project PDF report/documentation. • Interpret real-world policy impact of partials.

3 Spreadsheet Data Requirements (`flood_data.xlsx`)

Member 2 will compile historical flood event rows with the following explicit column headers:

1. **Region:** Name/Zone of the flood-affected region.
2. **PopulationDensity_x:** Estimated population density (x).
3. **EvacuationRoutes_y:** Number of open/functional evacuation routes (y).
4. **RescuedCount:** Total number of individuals evacuated/rescued.
5. **EvacuationTime_hrs:** Recorded total duration of evacuation (hours).

4 Proposed Project Report Structure

1. **Title Page:** Title, Team Members, Subject/Club details.
2. **Introduction & Social Impact:** Problem framing and the logistics of flood evacuation.
3. **Mathematical Formulation:** Definition of $U(x, y)$, $\frac{\partial U}{\partial x}$ (density sensitivity), and $\frac{\partial U}{\partial y}$ (route sensitivity).
4. **Data Collection:** Overview of historical flood data collected in Excel.
5. **MATLAB Analysis & Results:** Code execution output, derivative values per region, and sensitivity interpretation.
6. **Conclusion:** Key takeaways for disaster management teams.